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From Averages to Atavism: How the Victorian Rejection of Variation Shaped Early Criminal Profiling

Many people believe that science is a straight path toward a neutral truth, unaffected by personal opinions or the surrounding culture. However, looking back at history shows a much messier reality. During the Victorian era, there was an obsession with putting things in order and using numbers to explain human behavior. This period shows that scientific work often mirrors what a society wants to believe rather than how the world actually works. Back then, a concept called biological determinism became very popular. This theory claimed that a person's value, intelligence, and even whether they might commit a crime were set at birth by their biology. The language used to support this idea focused on strict categories and mathematical averages while ignoring how unique individuals can be. By looking at the work of the late evolutionary biologist Stephen Jay Gould and more recent historical research, it is clear that the Victorian failure to account for these differences was not just a simple mistake. Instead, it was a way to use the language of science to make social prejudices look like facts.

Gould was a famous scientist and author who spent much of his career studying the history of these problematic ideas. In one of his writings, he calls himself a "Galilean" author, someone who finds true excitement in solving nature's puzzles and trying to understand the world (207). In his book *Bully for Brontosaurus*, he explains the difference between two ways of writing about science: the "Franciscan" and the "Galilean." The Franciscan style focuses on the beauty and poetry of nature. In contrast, the Victorians used the Galilean style to make their work seem perfectly objective. They did this by using lots of numbers, charts, and tables to present their theories as if they were unchangeable laws. Gould uses his own lively writing style to show that this objectivity is often just an act. He argues that science is actually a "creative human activity." Because of this, he warns that we must be careful when scientific findings seem to support existing social or political biases. He believes that science should be easy for regular people to understand because we need "active inquiring citizens" who can question experts when their data is used to justify treating people unfairly (213).

The idea of biological determinism grew out of a Victorian desire to sort people into neat, easy-to-manage groups. As the *Encyclopedia Britannica* explains, this theory suggests that human behavior is controlled by genes or other physical factors. During the 1800s, there was a massive effort to find the "average" for different groups of people to decide what was "normal." This was as much about social control as it was about science. Richard Lewontin, a prominent genetics professor at Harvard, described this as an "ideological weapon" used to defend the way society was set up. By claiming that the gap between the rich and the poor was based on biology, those in power could argue that they were just following the laws of nature. By focusing on "types," the Victorians ignored the complexity of human beings. They claimed certain groups were biologically meant to be at the bottom of society. This went against the humanistic approach that Gould valued, which sees people as more than just a collection of physical parts (213).

Gould's best example of how people misused the idea of the average is found in his study of 19th-century craniometry, or brain science. He looked closely at the work of Paul Broca, a

French doctor who claimed that women were less intelligent because their brains weighed less on average. Broca assumed that a smaller body meant a smaller mind. Researchers Stanley Finger and Paul Eling point out in their historical analysis that Victorian scientists were always looking for a "veneer of mathematics" to support the ideas they already held (145). By focusing only on one number—the average—Broca ignored the huge variety among individual people. He took a single statistic and turned it into a biological rule. Gould calls this a rhetorical trick used to stop people from arguing by using the authority of numbers.

This habit of ignoring individual differences led to tragic results in the legal system. Instead of looking at why people committed crimes—like being poor or living in crowded cities—the Victorians looked for a biological cause. They wanted to find a specific "criminal type." They believed you could tell if someone was a criminal by measuring their skull, jaw, or arms. This theory was pushed by Cesare Lombroso, an Italian physician often called the "father of modern criminology." He argued that criminals were "atavists," or evolutionary throwbacks who had the traits of primitive humans. To Lombroso, a criminal was simply born the wrong way. He used the Victorian focus on categories to create a rigid profile of the "natural born criminal." He looked for specific physical marks, like wide jaws or large ears, which he called "stigmata." This turned police work into a study of biology. The logic was simple: if you looked like the average criminal type, you were destined to be one. This allowed the system to label people as biological threats rather than looking at their actual intent or actions. Gould notes that this turned people into "labels" instead of individuals (209).

These techniques were also closely tied to ideas about race. Scholars W. Carson Byrd and Matthew W. Hughey argue in their research on racial inequality that the Victorian focus on sorting people led to an "ideological double helix" of lasting inequality. By treating race as a biological category with its own set of averages, 19th-century science allowed racism to be built into the foundation of investigative work. This language helped the ruling class maintain control while claiming their methods were neutral. Historian Garland Edward Allen notes that this helped keep the "social and economic status quo" by suggesting that people at the bottom of society were there because of their genes rather than an unfair system.

Gould's strongest argument against this logic comes from his personal life. After being diagnosed with a serious illness, he realized that the average survival time he was given was just a simple number that didn't account for his own unique body. This led him to famously say that "variation itself is nature's only irreducible essence" (214). He argued that while an average can describe the middle point of a big group, it can't tell you anything about a specific person. In the Victorian era, the average was seen as the most important part of the story, and anyone who was different was seen as a mistake. Gould shows that the differences are what's real, while the average is just something humans made up to try to organize a complicated world.

Gould follows this same line of thinking in his "Prologue" to *Bully for Brontosaurus*. He defends his choice to connect science with politics, culture, and history, arguing that science cannot be separated from the society that produces it. He wants his readers to be "active inquiring citizens" who are willing to question the experts (213). By pointing out the problems in the Victorian view of the world, Gould shows that when we let experts define the average, we give them the power to decide what we are worth as humans (215). He explores this further by looking at the 19th-century naturalist Louis Agassiz. Even though Agassiz was a brilliant man, his belief in fixed categories caused him to deny that humans are equal. This shows that even the most objective scientists can be blinded by their own culture (215-221).

To fully understand this problem, we must see how Victorian ideas still stick around today. We don't measure skulls anymore, but we still make the mistake of sorting people into fixed types using data and algorithms. Using numbers to predict how people will behave can lead to the same errors Lombroso made. When we use computer programs to decide who might commit a crime or who should get a job, we are often choosing a statistical average over a real person. As Lewontin suggests, the only way to stop science from being used as a tool for control is to constantly remind ourselves of how different every person is. Ignoring these differences was a trick used by Victorians to avoid the complicated truth that all humans are equal.

Ultimately, the Victorian style of using numbers was a way to make themselves seem more credible, establishing what rhetoricians call "ethos." They used charts and figures to look invincible and logical. On the other hand, Gould uses "pathos," or personal stories, to connect with the emotions of his readers. He encourages people to "rage mightily against the dying of the light" (214). By doing this, he closes the gap between the scientist and the regular citizen. He sees them as the same, focusing on their shared humanity rather than a statistical category. As he notes in *Evolution as Fact and Theory*, scientific truth is never finished; it is a constant process of "checking and correcting" (231). The Victorians wanted their categories to be permanent. They wanted a standard criminal, a standard woman, and a standard worker. But nature doesn't work that way. Evolution depends on variety. Without it, there is no way for life to change or adapt. So, the Victorian denial of these differences was not just a social mistake; it was scientifically wrong. The focus on averages meant people were judged by the shape of their heads rather than their character. As Gould explains, the real truth of nature isn't found in a single middle point, but in the huge and beautiful variety of human life.

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